

CAPABILITIES

JTAG and scan

- Primary TAP with IDCODE and BYPASS; SAMPLE/PRELOAD and EXTEST are compile-time optional with boundary scan enabled
- Compile-time optional INTEST, CLAMP, HIGHZ, RUNBIST, EXTEST_PULSE, EXTEST_TRAIN, TMP, and CLAMP hold/release functions
- Boundary-scan host interface plus separate DFD, secure DFT, non-secure DFT, extended STAP, I/O STAP, SMC STAP, SEP STAP, and extra-STAP connections
- Capability registers report enabled functions and JTAG-to-AXI limits

Debug and control

- Single and series JTAG-to-AXI transactions, including incrementing and fixed-address series operations and target error reporting
- Independent lifecycle disable fields for three bridge paths and eight scan-side paths
- Independently enabled reset-control outputs for SMC, SEP, and adopter logic
- Debugger-controlled boot-stall and functional clock-stop requests

Cross trigger

- Bidirectional external CTP signaling and chiplet-local trigger interfaces
- Software-controlled source/destination routing with point-to-point and wire-OR behavior
- Pulse synchronization or four-phase request/acknowledge mode per internal trigger interface

INTERFACES AND CONFIGURATION

Feature	Integration options
Clocks and resets	System clock/reset, JTAG TCK/TRST_N, JTAG power-on reset, scan reset, test enable
Lifecycle policy	Active-high, per-path debug and test disable controls
Scan	PTAP pins; BSR/iJTAG scan controls; I/O, SMC, SEP, and extra STAPs
Fabric	One AXI4 manager, two AXI4-Lite managers, one AXI4-Lite subordinate
Cross trigger	External pad controls, internal source/sink interfaces, clock-stop requests and output
Build-time	Instruction/path enables, extra STAP count, IDCODE/version fields, reset-control output types, trigger protocol modes, AXI type parameters

DOCUMENT CONTROL

Version	Date	Author	Change

INTEGRATION DEPENDENCIES

- Generate lifecycle debug-disable policy outside DTP and connect the current active-high debug-disable contract.
- Replace the zero-valued IDCODE manufacturer, part, revision, and OCAH version defaults with release-specific values.
- Provide safe clock, reset, CDC, pad, and clock-gating integration, and establish project-specific TCK constraints.
- Configure the SMC, SEP, and adopter reset-control output types and enables, and connect or correctly bypass every enabled scan and secondary-TAP chain.
- Connect enabled AXI manager ports only to address maps permitted by the product security policy. Disabled and unused interfaces require documented tie-offs.
- Integrate the cross-trigger CSR subordinate into the management address map and connect the required local, external, and clock-stop endpoints.

VERIFICATION AND MATURITY

The OCAH [verification dashboard](#) publishes the current verification and maturity status. The repository’s open-source PyUVM environment exercises TAP operation, standard and optional instructions, JTAG-to-AXI data and error paths, backpressure/reset cases, iJTAG and STAP routing, cross-trigger modes, clock controls, and lifecycle-disable matrices. Scoreboards, reference models, coverage intent, and deliberately corrupted-checker tests guard against vacuous passes.

This beta sheet does not specify a characterized operating frequency, and no third-party standards certification, silicon characterization, or portable frequency, area, and power figures are claimed. Those results depend on the selected configuration, libraries, constraints, physical implementation, and target process.

DELIVERABLES

- SystemVerilog RTL and dependency filelists
- SystemRDL-derived cross-trigger register collateral
- Integration, architecture, interface, and verification-plan documentation
- Open-source cocoth/PyUVM tests, reusable JTAG/AXI agents, scoreboards, reference models, and functional coverage
- Lint collateral and lint, CDC, and RDC waivers
- Synthesis and timing-constraint starting points
- Apache License 2.0 repository distribution

RESOURCES

Documentation: [tenstorrent.github.io/tt-oca-harness](#)
DTP source documentation: [hw/sys/dtp/doc/](#)
RTL: [hw/sys/dtp/rtl/](#)
Verification: [hw/sys/dtp/dv/](#)
Synthesis collateral: [hw/sys/dtp/synth/](#)
Integrator Guide: [doc/integrator/](#)
Project: [github.com/tenstorrent/tt-oca-harness](#)

TERMS

BSR: boundary-scan register. CSR: control and status register. CTP: cross trigger port. DFD: design-for-debug. DFT: design for test. iJTAG: internal JTAG (IEEE 1687). OTP: one-time programmable. PTAP: primary TAP. SEP: security processor. SMC: system management controller. SMU: system management unit. STAP: secondary TAP. TMP: test mode persistence.